

Benchmark Comparison with Standard Longitudinal Models

1 Benchmark comparison with standard longitudinal models

We compared the default `gamlss.longitudinal` RS-joint fit against `glm` and `geepack` GEE baselines across Gaussian, Gamma, binary, and Poisson longitudinal simulations. The GEE baselines used exchangeable, AR(1), and unstructured working correlation. GAM and GLMM rows were removed from this report because GAM duplicated the fixed-effect `glm` baseline here, while GLMM targets a conditional mixed-model estimand that is less directly comparable for this marginal response-mean benchmark. The Poisson results are complete for all but the covariate-dependent high-correlation setting, where 51 replicates were available.

Each simulation used four repeated observations per subject and the same fixed-effect marginal mean formula, response $\sim x + z_{null}$. The covariate x was the active slope used for coefficient calibration, while z_{null} was included as a true-null covariate for false-positive-rate checks. The benchmark deliberately excludes low-correlation scenarios and focuses on moderate to high within-subject dependence.

Table 1: Simulation scenarios included in the benchmark. External scenarios were generated independently of `gamlss.longitudinal` using a Gaussian copula and fixed correlation matrix. Internal scenarios used the package `longitudinal` simulator to vary the adjacent dependence parameter over time or by covariate value.

Scenario	Simulator	True dependence	Level	Detail
External exchangeable, moderate	external	exchangeable	moderate	Gaussian copula, rho=0.45
External exchangeable, high	external	exchangeable	high	Gaussian copula, rho=0.75
External AR(1), moderate	external	ar1	moderate	Gaussian copula, rho=0.55
External AR(1), high	external	ar1	high	Gaussian copula, rho=0.78
Internal time-varying adjacent	internal	time_varying_adjacent	moderate.high	Gaussian copula, adjacent tau=0.35, 0.55, 0.70
Internal covariate-dependent adjacent	internal	covariate_dependent_adjacent	moderate.high	Gaussian copula, adjacent tau=inverse-logit(logit(0.55)+1.1x)

Table 2: Median response mean RMSE by simulation scenario and model. Lower values are better. These marginal mean metrics are expected to be relatively similar across models because all fitted models use the same fixed-effect mean formula.

Scenario	<code>gamlss.longitudinal</code>	<code>geepack</code> exch.	<code>geepack</code> AR(1)	<code>geepack</code> unstr.	<code>glm</code>
External exchangeable, moderate	0.089	0.090	0.089	0.101	0.090
External exchangeable, high	0.104	0.108	0.106	0.127	0.108
External AR(1), moderate	0.089	0.092	0.089	0.096	0.092
External AR(1), high	0.099	0.104	0.102	0.118	0.104
Internal time-varying adjacent	0.102	0.103	0.102	0.111	0.103
Internal covariate-dependent adjacent	0.099	0.099	0.098	0.109	0.099

The revised comparison focuses on marginal exponential-family baselines. The scenario-wise tables are the primary comparison because the dependence structure, rather than the response family, is the main experimental contrast. The `glm` rows show the cost of ignoring within-subject correlation for inference, while the `geepack` rows show how much can be recovered with standard working-correlation choices. The unstructured `geepack` row is included as a flexible working-correlation comparator; 2349 of 2351 unstructured GEE fits completed successfully, with two timeout/failure rows retained as failed benchmark attempts.

Table 3: Mean slope standard-error ratio by simulation scenario and model. Values closer to one indicate better calibration of the model-based standard error against the empirical Monte Carlo standard deviation.

Scenario	gamlss.longitudinal	geepack exch.	geepack AR(1)	geepack unstr.	glm
External exchangeable, moderate	0.963	1.070	1.076	0.978	0.726
External exchangeable, high	0.889	0.971	0.983	0.883	0.559
External AR(1), moderate	0.994	0.989	0.994	1.008	0.698
External AR(1), high	1.030	1.030	1.035	0.506	0.616
Internal time-varying adjacent	0.941	0.955	0.951	0.920	0.598
Internal covariate-dependent adjacent	0.983	1.002	0.990	1.005	0.612

Table 4: Mean empirical 95% confidence interval coverage for the slope coefficient by simulation scenario and model. Values closer to 0.95 are better.

Scenario	gamlss.longitudinal	geepack exch.	geepack AR(1)	geepack unstr.	glm
External exchangeable, moderate	0.927	0.948	0.955	0.953	0.835
External exchangeable, high	0.913	0.932	0.925	0.910	0.735
External AR(1), moderate	0.948	0.948	0.953	0.952	0.823
External AR(1), high	0.945	0.955	0.948	0.932	0.758
Internal time-varying adjacent	0.938	0.930	0.948	0.940	0.745
Internal covariate-dependent adjacent	0.930	0.943	0.940	0.950	0.739

Table 5: Mean null false-positive rate by simulation scenario and model. Lower values are better, with nominal two-sided 5% testing targeting 0.05.

Scenario	gamlss.longitudinal	geepack exch.	geepack AR(1)	geepack unstr.	glm
External exchangeable, moderate	0.075	0.037	0.050	0.042	0.185
External exchangeable, high	0.082	0.070	0.075	0.070	0.292
External AR(1), moderate	0.070	0.077	0.072	0.065	0.200
External AR(1), high	0.050	0.050	0.055	0.065	0.263
Internal time-varying adjacent	0.048	0.055	0.062	0.073	0.263
Internal covariate-dependent adjacent	0.052	0.062	0.055	0.032	0.226

Table 6: Number of scenario-family cells in which each model had the best value for each benchmark metric. Larger counts indicate broader dominance across the simulation grid.

Metric	gamlss.longitudinal	geepack exch.	geepack AR(1)	geepack unstr.	glm
Response mean RMSE	10	0	8	0	6
90th percentile MAE	20	10	7	7	9
95% prediction interval coverage	19	20	20	18	20
Upper-tail error	7	3	7	7	0
Runtime	0	0	0	0	24

Table 7: Median response mean RMSE by response family and model. Lower values are better.

Family	gamlss.longitudinal	geepack exch.	geepack AR(1)	geepack unstr.	glm
Gaussian	0.106	0.108	0.105	0.119	0.108
Gamma	0.093	0.095	0.094	0.102	0.095
Binary	0.045	0.046	0.046	0.052	0.046
Poisson	0.193	0.189	0.191	0.212	0.189

Table 8: Average coefficient-level inference calibration across the benchmark grid. The SE ratio compares the model-based standard error with the empirical Monte Carlo standard deviation for the slope coefficient; values near 1 are best. CI coverage is the empirical coverage of nominal 95% intervals for the slope coefficient. The false-positive rate is computed for the null coefficient.

Method	SE ratio	95% CI coverage	Null false-positive rate
gamlss.longitudinal	0.967	0.933	0.063
geepack exchangeable	1.003	0.943	0.059
geepack AR(1)	1.005	0.945	0.062
geepack unstructured	0.883	0.940	0.058
glm	0.635	0.772	0.238

Table 9: Median elapsed time in seconds by response family and model. Lower values are faster.

Family	gamlss.longitudinal	geepack exch.	geepack AR(1)	geepack unstr.	glm
Gaussian	3.02	0.02	0.02	0.02	0.00
Gamma	4.08	0.02	0.03	0.02	0.01
Binary	16.34	0.02	0.03	0.02	0.01
Poisson	52.73	0.01	0.02	0.02	0.00

A Metric-by-metric review tables

The first appendix block repeats the scenario-by-model layout for every numeric metric in the saved review table. The second block keeps the family-by-model summaries as a secondary response-family check.

A.1 Scenario-by-model summaries

Table 10: Median response-mean RMSE by simulation scenario and model. Lower values are better.

Scenario	gamlss.longitudinal	geepack exch.	geepack AR(1)	geepack unstr.	glm
External exchangeable, moderate	0.089	0.090	0.089	0.101	0.090
External exchangeable, high	0.104	0.108	0.106	0.127	0.108
External AR(1), moderate	0.089	0.092	0.089	0.096	0.092
External AR(1), high	0.099	0.104	0.102	0.118	0.104
Internal time-varying adjacent	0.102	0.103	0.102	0.111	0.103
Internal covariate-dependent adjacent	0.099	0.099	0.098	0.109	0.099

Table 11: Median 90th percentile MAE by simulation scenario and model. Lower values are better.

Scenario	gamlss.longitudinal	geepack exch.	geepack AR(1)	geepack unstr.	glm
External exchangeable, moderate	0.101	0.104	0.104	0.125	0.104
External exchangeable, high	0.125	0.129	0.129	0.141	0.129
External AR(1), moderate	0.105	0.108	0.107	0.112	0.108
External AR(1), high	0.113	0.121	0.118	0.140	0.121
Internal time-varying adjacent	0.119	0.124	0.120	0.130	0.124
Internal covariate-dependent adjacent	0.112	0.116	0.116	0.130	0.116

Table 12: Median upper-tail probability error at the true 90th percentile by simulation scenario and model. Values closer to zero are better.

Scenario	gamlss.longitudinal	geepack exch.	geepack AR(1)	geepack unstr.	glm
External exchangeable, moderate	-0.000	-0.001	-0.000	-0.002	-0.001
External exchangeable, high	-0.000	-0.000	0.000	-0.002	-0.000
External AR(1), moderate	0.000	-0.000	-0.000	0.000	-0.000
External AR(1), high	-0.001	-0.002	-0.002	-0.001	-0.002
Internal time-varying adjacent	-0.001	-0.002	-0.002	-0.001	-0.002
Internal covariate-dependent adjacent	-0.000	-0.001	-0.000	0.001	-0.001

Table 13: Median empirical 95% prediction interval coverage by simulation scenario and model. Values closer to 0.95 are better.

Scenario	gamlss.longitudinal	geepack exch.	geepack AR(1)	geepack unstr.	glm
External exchangeable, moderate	0.968	0.967	0.967	0.967	0.967
External exchangeable, high	0.968	0.968	0.968	0.969	0.968
External AR(1), moderate	0.968	0.968	0.968	0.967	0.968
External AR(1), high	0.968	0.968	0.967	0.968	0.968
Internal time-varying adjacent	0.968	0.968	0.968	0.969	0.968
Internal covariate-dependent adjacent	0.968	0.968	0.968	0.966	0.968

Table 14: Median negative log score by simulation scenario and model. Lower values are better.

Scenario	gamlss.longitudinal	geepack exch.	geepack AR(1)	geepack unstr.	glm
External exchangeable, moderate	1.324	1.324	1.324	1.320	1.324
External exchangeable, high	1.319	1.319	1.319	1.316	1.319
External AR(1), moderate	1.331	1.331	1.332	1.330	1.331
External AR(1), high	1.321	1.320	1.321	1.325	1.320
Internal time-varying adjacent	1.320	1.319	1.319	1.319	1.319
Internal covariate-dependent adjacent	1.324	1.324	1.324	1.324	1.324

Table 15: Median elapsed time in seconds by simulation scenario and model. Lower values are faster.

Scenario	gamlss.longitudinal	geepack exch.	geepack AR(1)	geepack unstr.	glm
External exchangeable, moderate	4.56	0.01	0.02	0.02	0.00
External exchangeable, high	9.40	0.02	0.02	0.02	0.00
External AR(1), moderate	8.33	0.02	0.02	0.02	0.00
External AR(1), high	11.02	0.02	0.03	0.02	0.01
Internal time-varying adjacent	29.05	0.03	0.03	0.02	0.01
Internal covariate-dependent adjacent	22.77	0.05	0.05	0.02	0.01

Table 16: Median number of replicates contributing to each response family and model summary.

Scenario	gamlss.longitudinal	geepack exch.	geepack AR(1)	geepack unstr.	glm
External exchangeable, moderate	100	100	100	100	100
External exchangeable, high	100	100	100	100	100
External AR(1), moderate	100	100	100	100	100
External AR(1), high	100	100	100	100	100
Internal time-varying adjacent	100	100	100	100	100
Internal covariate-dependent adjacent	100	100	100	100	100

Table 17: Mean slope standard-error ratio by simulation scenario and model. Values closer to one are better.

Scenario	gamlss.longitudinal	geepack exch.	geepack AR(1)	geepack unstr.	glm
External exchangeable, moderate	0.963	1.070	1.076	0.978	0.726
External exchangeable, high	0.889	0.971	0.983	0.883	0.559
External AR(1), moderate	0.994	0.989	0.994	1.008	0.698
External AR(1), high	1.030	1.030	1.035	0.506	0.616
Internal time-varying adjacent	0.941	0.955	0.951	0.920	0.598
Internal covariate-dependent adjacent	0.983	1.002	0.990	1.005	0.612

Table 18: Mean empirical 95% confidence interval coverage for the slope coefficient by simulation scenario and model. Values closer to 0.95 are better.

Scenario	gamlss.longitudinal	geepack exch.	geepack AR(1)	geepack unstr.	glm
External exchangeable, moderate	0.927	0.948	0.955	0.953	0.835
External exchangeable, high	0.913	0.932	0.925	0.910	0.735
External AR(1), moderate	0.948	0.948	0.953	0.952	0.823
External AR(1), high	0.945	0.955	0.948	0.932	0.758
Internal time-varying adjacent	0.938	0.930	0.948	0.940	0.745
Internal covariate-dependent adjacent	0.930	0.943	0.940	0.950	0.739

Table 19: Mean null-effect standard-error ratio by simulation scenario and model. Values closer to one are better.

Scenario	gamlss.longitudinal	geepack exch.	geepack AR(1)	geepack unstr.	glm
External exchangeable, moderate	0.936	1.036	1.039	1.028	0.704
External exchangeable, high	0.873	0.933	0.936	0.935	0.537
External AR(1), moderate	0.972	0.955	0.954	0.947	0.669
External AR(1), high	0.982	0.978	0.977	0.608	0.586
Internal time-varying adjacent	0.956	0.955	0.951	0.833	0.594
Internal covariate-dependent adjacent	0.997	0.986	0.987	1.042	0.596

Table 20: Mean empirical 95% confidence interval coverage for the null coefficient by simulation scenario and model. Values closer to 0.95 are better.

Scenario	gamlss.longitudinal	geepack exch.	geepack AR(1)	geepack unstr.	glm
External exchangeable, moderate	0.925	0.962	0.950	0.958	0.815
External exchangeable, high	0.917	0.930	0.925	0.930	0.708
External AR(1), moderate	0.930	0.922	0.927	0.935	0.800
External AR(1), high	0.950	0.950	0.945	0.935	0.738
Internal time-varying adjacent	0.952	0.945	0.938	0.927	0.738
Internal covariate-dependent adjacent	0.948	0.938	0.945	0.968	0.774

Table 21: Mean null false-positive rate by simulation scenario and model. Lower values are better, with nominal two-sided 5% testing targeting 0.05.

Scenario	gamlss.longitudinal	geepack exch.	geepack AR(1)	geepack unstr.	glm
External exchangeable, moderate	0.075	0.037	0.050	0.042	0.185
External exchangeable, high	0.082	0.070	0.075	0.070	0.292
External AR(1), moderate	0.070	0.077	0.072	0.065	0.200
External AR(1), high	0.050	0.050	0.055	0.065	0.263
Internal time-varying adjacent	0.048	0.055	0.062	0.073	0.263
Internal covariate-dependent adjacent	0.052	0.062	0.055	0.032	0.226

Table 22: Mean gamlss.longitudinal primary fit success rate attached to each response family and model row.

Scenario	gamlss.longitudinal	geepack exch.	geepack AR(1)	geepack unstr.	glm
External exchangeable, moderate	1.000	1.000	1.000	1.000	1.000
External exchangeable, high	1.000	1.000	1.000	1.000	1.000
External AR(1), moderate	1.000	1.000	1.000	1.000	1.000
External AR(1), high	1.000	1.000	1.000	1.000	1.000
Internal time-varying adjacent	1.000	1.000	1.000	1.000	1.000
Internal covariate-dependent adjacent	1.000	1.000	1.000	1.000	1.000

Table 23: Mean gamlss.longitudinal primary fit convergence rate attached to each response family and model row.

Scenario	gamlss.longitudinal	geepack exch.	geepack AR(1)	geepack unstr.	glm
External exchangeable, moderate	1.000	1.000	1.000	1.000	1.000
External exchangeable, high	1.000	1.000	1.000	1.000	1.000
External AR(1), moderate	1.000	1.000	1.000	1.000	1.000
External AR(1), high	1.000	1.000	1.000	1.000	1.000
Internal time-varying adjacent	1.000	1.000	1.000	1.000	1.000
Internal covariate-dependent adjacent	1.000	1.000	1.000	1.000	1.000

Table 24: Median gamlss.longitudinal primary fit elapsed time in seconds attached to each response family and model row.

Scenario	gamlss.longitudinal	geepack exch.	geepack AR(1)	geepack unstr.	glm
External exchangeable, moderate	4.56	4.56	4.56	4.56	4.56
External exchangeable, high	9.40	9.40	9.40	9.40	9.40
External AR(1), moderate	8.33	8.33	8.33	8.33	8.33
External AR(1), high	11.02	11.02	11.02	11.02	11.02
Internal time-varying adjacent	29.05	29.05	29.05	29.05	29.05
Internal covariate-dependent adjacent	22.77	22.77	22.77	22.77	22.77

A.2 Family-by-model summaries

Table 25: Median response-mean RMSE by response family and model. Lower values are better.

Family	gamlss.longitudinal	geepack exch.	geepack AR(1)	geepack unstr.	glm
Gaussian	0.106	0.108	0.105	0.119	0.108
Gamma	0.093	0.095	0.094	0.102	0.095
Binary	0.045	0.046	0.046	0.052	0.046
Poisson	0.193	0.189	0.191	0.212	0.189

Table 26: Median 90th percentile MAE by response family and model. Lower values are better.

Family	gamlss.longitudinal	geepack exch.	geepack AR(1)	geepack unstr.	glm
Gaussian	0.097	0.100	0.098	0.112	0.100
Gamma	0.127	0.136	0.134	0.151	0.136
Binary	0.000	0.000	0.000	0.000	0.000
Poisson	0.183	0.200	0.205	0.225	0.200

Table 27: Median upper-tail probability error at the true 90th percentile by response family and model. Values closer to zero are better.

Family	gamlss.longitudinal	geepack exch.	geepack AR(1)	geepack unstr.	glm
Gaussian	-0.002	-0.002	-0.001	-0.002	-0.002
Gamma	-0.001	-0.001	-0.001	-0.002	-0.001
Binary	0.000	—	—	—	—
Poisson	0.001	0.001	0.001	0.001	0.001

Table 28: Median empirical 95% prediction interval coverage by response family and model. Values closer to 0.95 are better.

Family	gamlss.longitudinal	geepack exch.	geepack AR(1)	geepack unstr.	glm
Gaussian	0.950	0.950	0.950	0.950	0.950
Gamma	0.950	0.950	0.950	0.950	0.950
Binary	1.000	1.000	1.000	1.000	1.000
Poisson	0.985	0.985	0.985	0.985	0.985

Table 29: Median negative log score by response family and model. Lower values are better.

Family	gamlss.longitudinal	geepack exch.	geepack AR(1)	geepack unstr.	glm
Gaussian	1.412	1.412	1.412	1.411	1.412
Gamma	1.232	1.232	1.232	1.232	1.232
Binary	0.638	0.638	0.639	0.639	0.638
Poisson	1.924	1.924	1.924	1.922	1.924

Table 30: Median elapsed time in seconds by response family and model. Lower values are faster.

Family	gamlss.longitudinal	geepack exch.	geepack AR(1)	geepack unstr.	glm
Gaussian	3.02	0.02	0.02	0.02	0.00
Gamma	4.08	0.02	0.03	0.02	0.01
Binary	16.34	0.02	0.03	0.02	0.01
Poisson	52.73	0.01	0.02	0.02	0.00

Table 31: Median number of replicates contributing to each response family and model summary.

Family	gamlss.longitudinal	geepack exch.	geepack AR(1)	geepack unstr.	glm
Gaussian	100	100	100	100	100
Gamma	100	100	100	100	100
Binary	100	100	100	100	100
Poisson	100	100	100	100	100

Table 32: Mean slope standard-error ratio by response family and model. Values closer to one are better.

Family	gamlss.longitudinal	geepack exch.	geepack AR(1)	geepack unstr.	glm
Gaussian	0.956	0.986	0.987	0.994	0.605
Gamma	0.997	1.026	1.033	0.830	0.629
Binary	0.934	0.982	0.988	0.917	0.672
Poisson	0.979	1.016	1.012	0.792	0.632

Table 33: Mean empirical 95% confidence interval coverage for the slope coefficient by response family and model. Values closer to 0.95 are better.

Family	gamlss.longitudinal	geepack excl.	geepack AR(1)	geepack unstr.	glm
Gaussian	0.930	0.935	0.932	0.952	0.757
Gamma	0.942	0.952	0.955	0.945	0.760
Binary	0.925	0.948	0.957	0.927	0.807
Poisson	0.937	0.935	0.935	0.935	0.766

Table 34: Mean null-effect standard-error ratio by response family and model. Values closer to one are better.

Family	gamlss.longitudinal	geepack excl.	geepack AR(1)	geepack unstr.	glm
Gaussian	1.001	1.028	1.033	1.013	0.629
Gamma	0.951	0.973	0.975	0.779	0.595
Binary	0.927	0.941	0.940	0.961	0.642
Poisson	0.931	0.953	0.948	0.842	0.592

Table 35: Mean empirical 95% confidence interval coverage for the null coefficient by response family and model. Values closer to 0.95 are better.

Family	gamlss.longitudinal	geepack excl.	geepack AR(1)	geepack unstr.	glm
Gaussian	0.945	0.950	0.950	0.933	0.778
Gamma	0.933	0.937	0.935	0.938	0.758
Binary	0.938	0.943	0.938	0.943	0.780
Poisson	0.932	0.935	0.930	0.953	0.731

Table 36: Mean null false-positive rate by response family and model. Lower values are better, with nominal two-sided 5% testing targeting 0.05.

Family	gamlss.longitudinal	geepack excl.	geepack AR(1)	geepack unstr.	glm
Gaussian	0.055	0.050	0.050	0.067	0.222
Gamma	0.067	0.063	0.065	0.062	0.242
Binary	0.062	0.057	0.062	0.057	0.220
Poisson	0.068	0.065	0.070	0.047	0.269

Table 37: Mean gamlss.longitudinal primary fit success rate attached to each response family and model row.

Family	gamlss.longitudinal	geepack excl.	geepack AR(1)	geepack unstr.	glm
Gaussian	1.000	1.000	1.000	1.000	1.000
Gamma	1.000	1.000	1.000	1.000	1.000
Binary	1.000	1.000	1.000	1.000	1.000
Poisson	1.000	1.000	1.000	1.000	1.000

Table 38: Mean gamlss.longitudinal primary fit convergence rate attached to each response family and model row.

Family	gamlss.longitudinal	geepack exch.	geepack AR(1)	geepack unstr.	glm
Gaussian	1.000	1.000	1.000	1.000	1.000
Gamma	1.000	1.000	1.000	1.000	1.000
Binary	1.000	1.000	1.000	1.000	1.000
Poisson	1.000	1.000	1.000	1.000	1.000

Table 39: Median gamlss.longitudinal primary fit elapsed time in seconds attached to each response family and model row.

Family	gamlss.longitudinal	geepack exch.	geepack AR(1)	geepack unstr.	glm
Gaussian	3.02	3.02	3.02	3.02	3.02
Gamma	4.08	4.08	4.08	4.08	4.08
Binary	16.34	16.34	16.34	16.34	16.34
Poisson	52.73	52.73	52.73	52.73	52.73